Word constituents and the morphology-syntax distinction: descriptive and typological perspectives II

Atelier Morphosyntaxe

Laboratoire Dynamique du Langage

CNRS, Lyon II

Adam J.R. Tallman

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Overview

- Review of last lecture
- Clitics as epiphenomena
- Two perspectives on wordhood tests or constituency diagnostics
- Review constituency diagnostics (but slower)

Review of last lecture

- Theories that distinguish between morphology and syntax make intuitively probabilistic claims about boundary cases or the interpretation of data.

- Theories that distinguish between morphology and syntax tend to lapse into circularity when faced with apparent counterevidence:
  - E.g. words are domains that cannot be interrupted, and domains that cannot be interrupted are words

Perspectives on the morphology-syntax distinction

- Taxis = its all about ordering!

- Relations = Its about head-dependent relations not providing clear results in the morphology.

- Biuniqueness = Its about morphological elements not always being biunique with respect to form and meaning

- Encapsulation = Its about syntax not having access to word-internal structure.

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- Encapsulation = Its about syntax not having access to word-internal structure.
  - No one knows how to draw word boundaries in a consistent fashion

“Words”

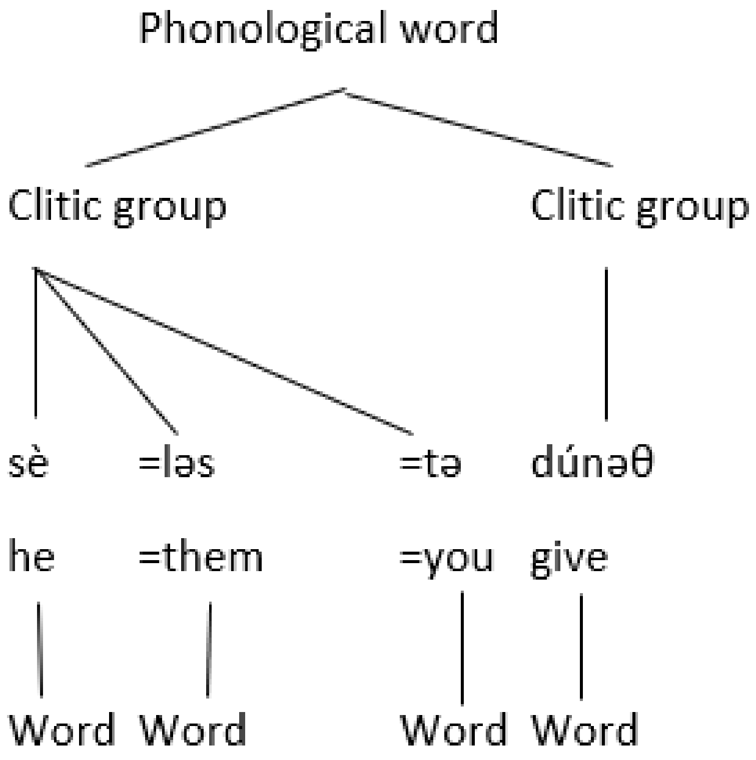
- At some point in the 1970s people decided to start saying that there was a distinction between morphosyntax / grammatical and phonological words.
    - Elordieta 2014

“Words”

- What is a phonological word?

- The phonological word is a domain of structure where some phonological process can be said to apply
  - E.g. stress assignment.

“Words”



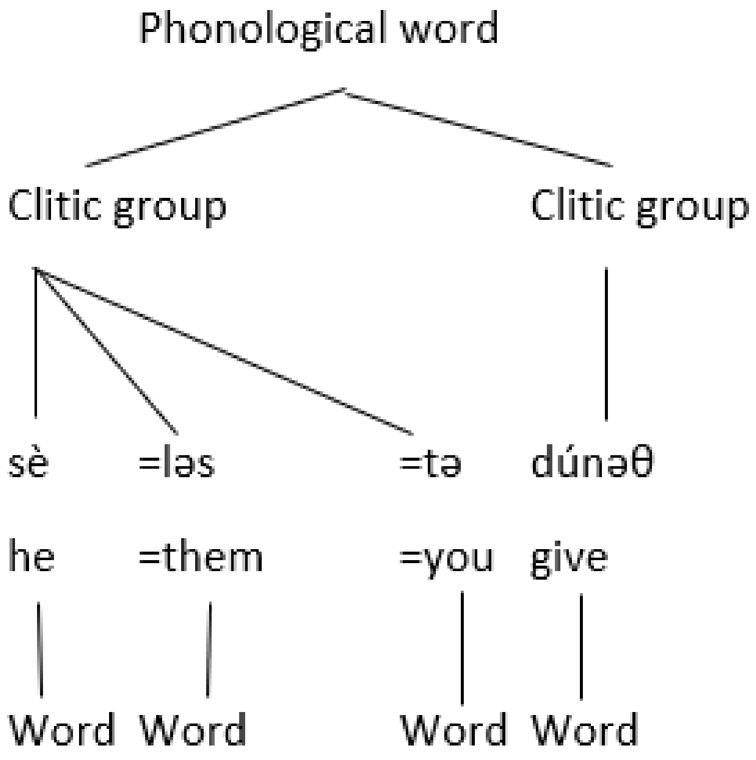
Old French: ‘If he gives them to you’ (Horne 1990)

“Words”

- What is a morphosyntactic word?

- The morphosyntactic word is some domain of structure where some constraint on ordering, interruptability applies that does not apply elsewhere in the grammar.

“Words”



Old French: ‘If he gives them to you’ (Horne 1990)

“Words” DISCLAIMER

- A lot of people determine word boundaries from orthographic practices.
- Maybe those orthographic practices have some linguistic basis, but should we really always assume this?
  - (think of spelling practice in English)
- Studies that look at languages where writing practices have only recently entered show that speakers are inconsistent with respect to word boundaries.
  - Pederson 2008; Tallman 2018

“Words”

- Anyways – the reason that a distinction between phonological and morphosyntactic words was made was to account for cases where the criteria do not line up.

- People found that there were phonological processes that applied to domains higher than what they found for the phonological word, and so the prosodic hierarchy was created.

- They made the prosodic hierarchy go all the way down to feet and syllables etc.

“Words”?

Utterance Phrase

Intonational Phrase

Phonological phrase

Phonological word

Foot

…

Words

- It has been assumed that this prosodic structure is another layer of phonological constituency that is non-isomorphic (fancy way of saying not the same while implying related to) with morphosyntactic constituency.

- And by the way …

“Words”?

Utterance Phrase

Intonational Phrase

Phonological phrase

Phonological word

Phonological word

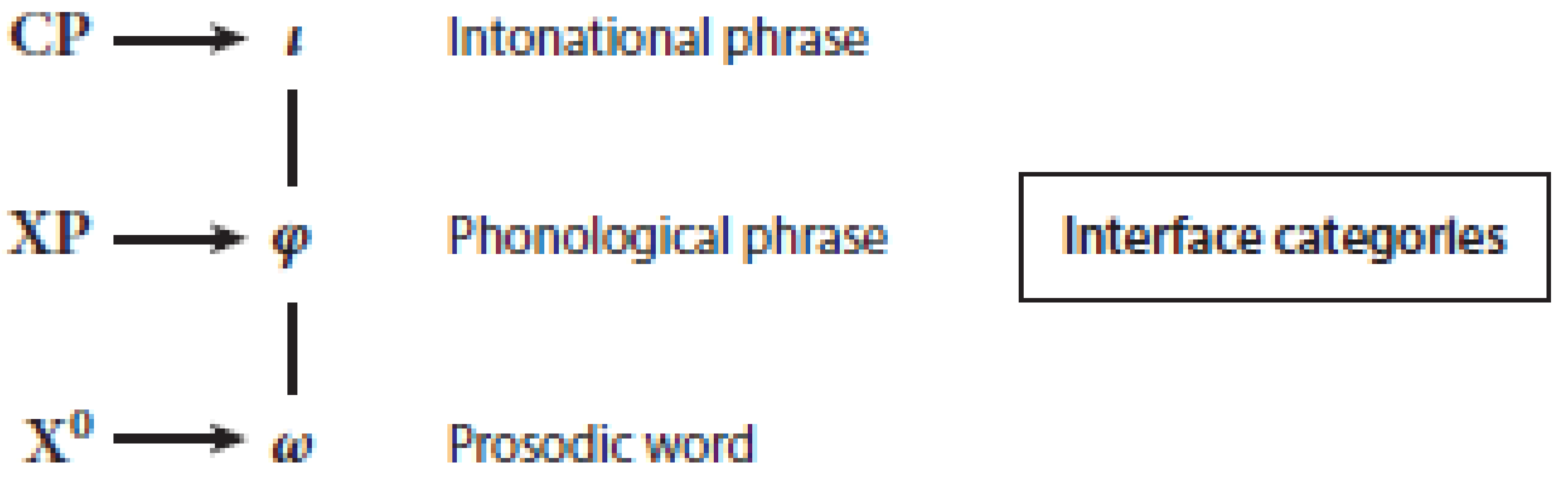
Irregular phonological rules

“Post-lexical phonology”
Stuff that happens
above words

“Lexical phonology”
Stuff that happens in
words

“Words”

- People who say they study the syntax-phonology interface study this stuff.
  - Bennet and Elfner 2019



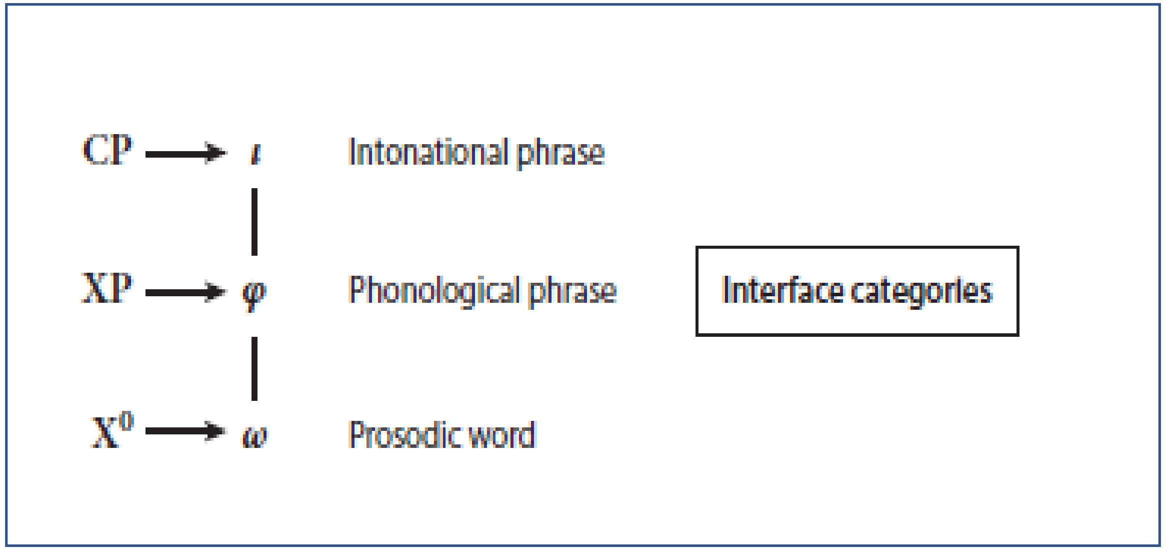
“Words”

- A lot of work has argued that phonological phrases can map onto morphosyntactic words.
  - (Rice 1993; Dyck 2009; Russell 1999; Evans et al. 2008; Ross 2011; Miller 2018)
- Or that prosodic words can map onto syntactic phrases.
  - Post (2009, 2017)

- Prosodic words can map onto affixes
  - Bickel et al. 2007; Woodbury 2011

- etc.

What generative grammarians want you to think



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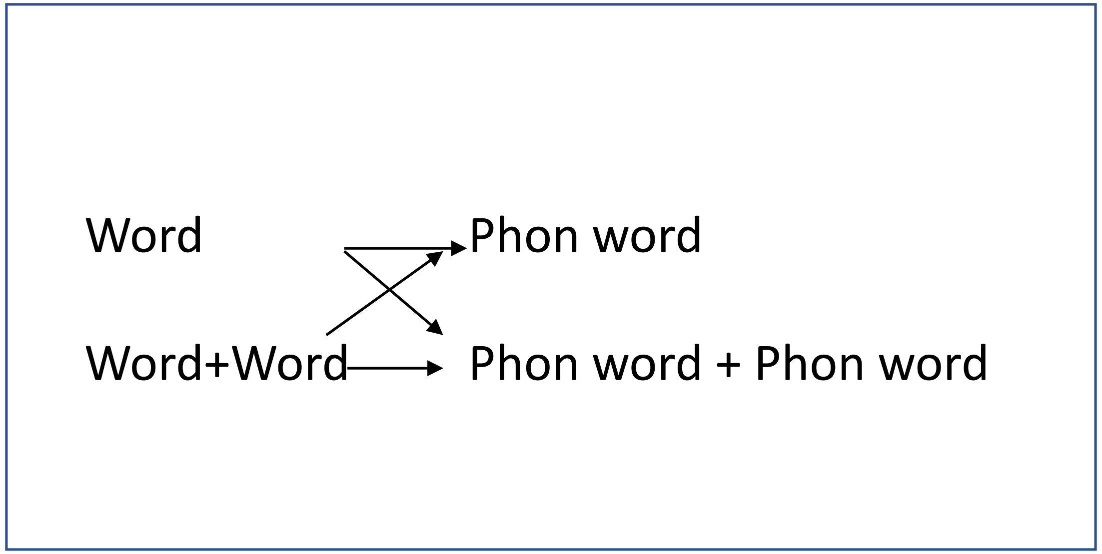

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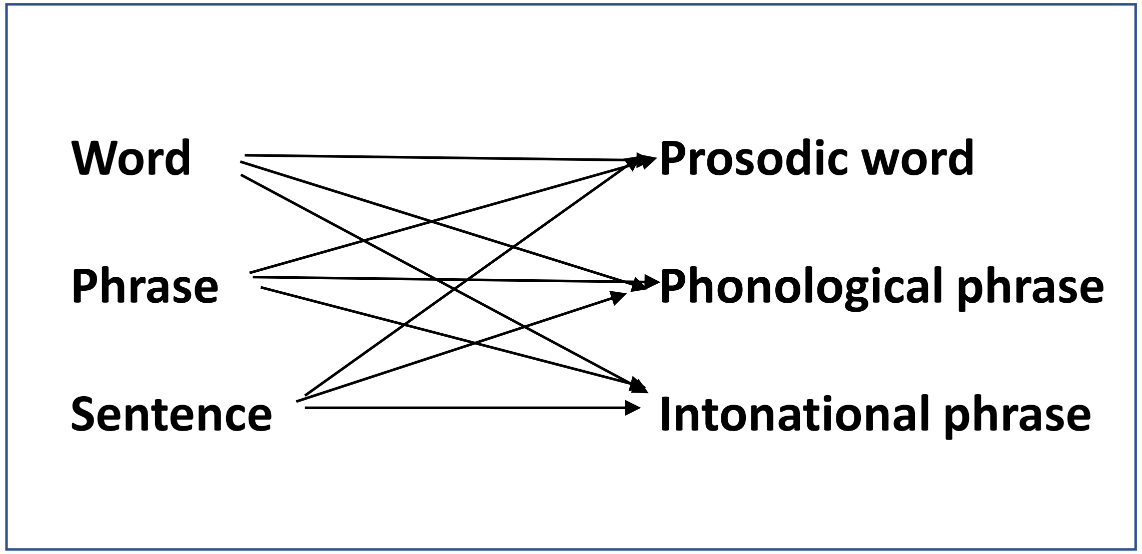

“Words”

- Can you think of any problems behind these proposals?

“Words”

- How do we know that a given category is a ‘word’ or any level whatsoever to begin with?
- There are no general rules in linguistic analysis for determining the level of any specific morphophonological process.
- And many languages either have too many or not enough domains motivated by morphophonological processes.

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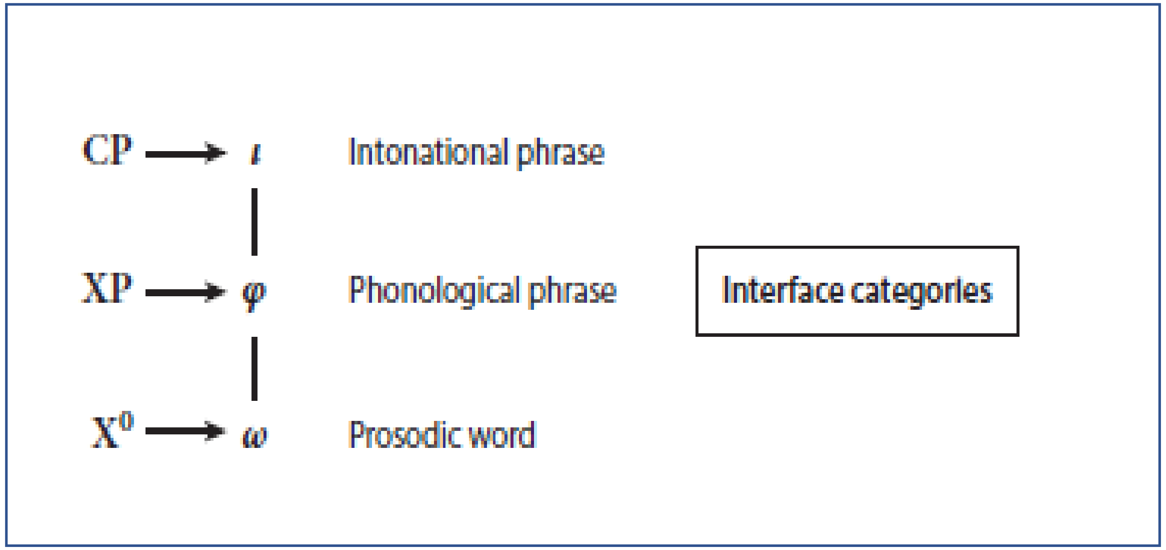

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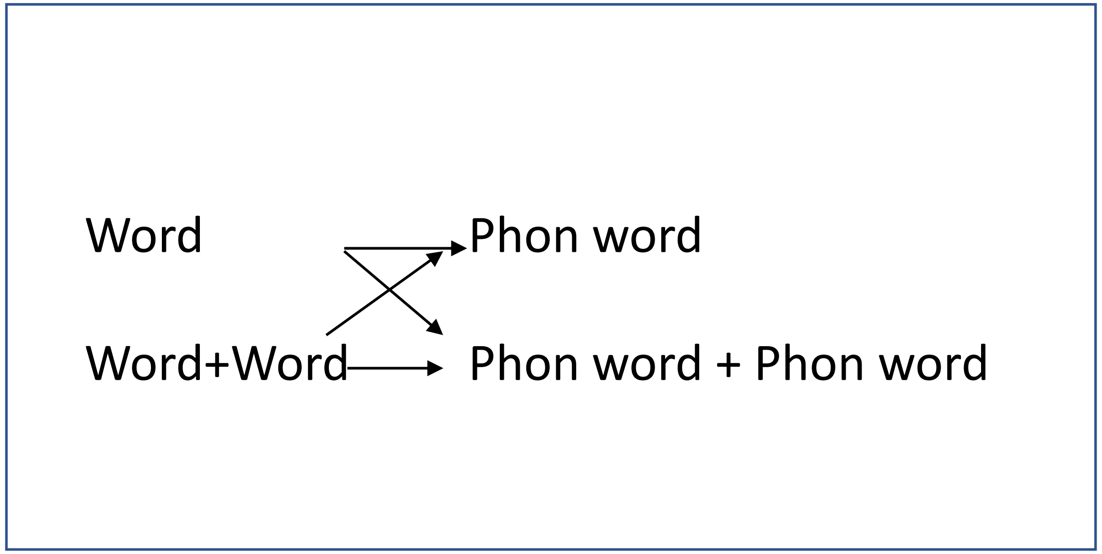

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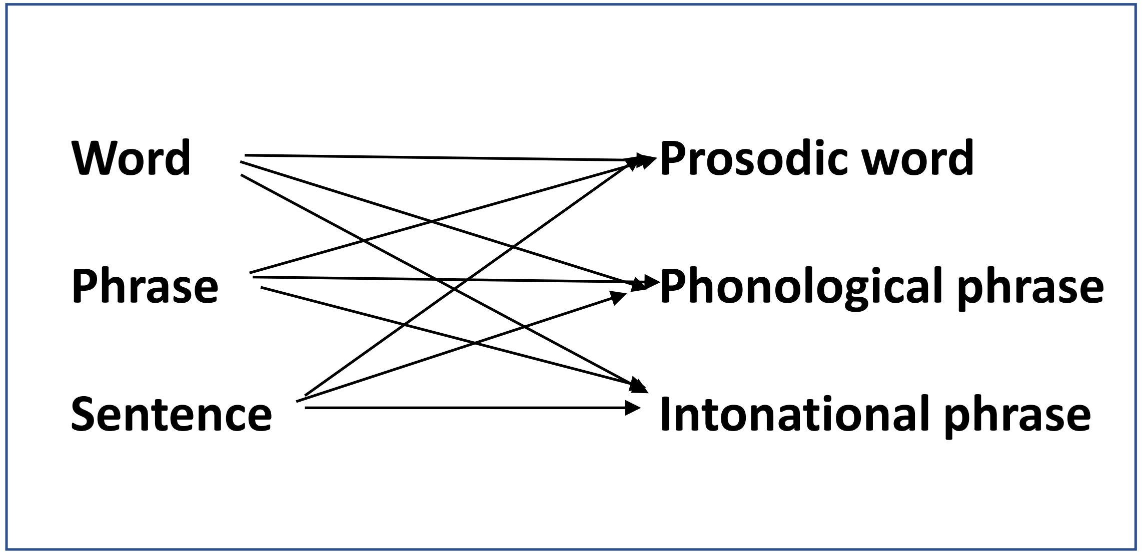

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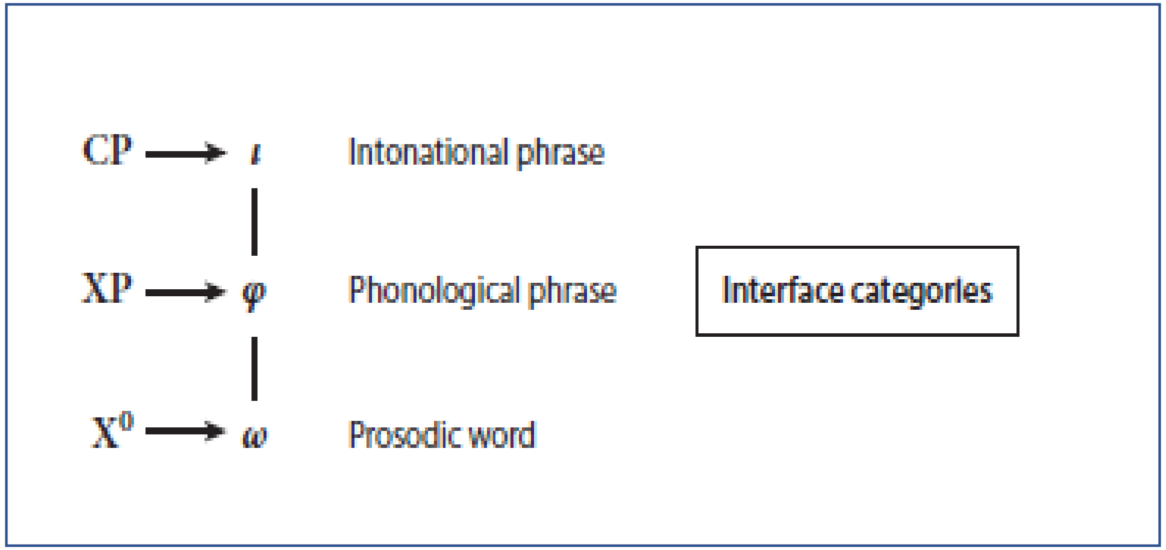

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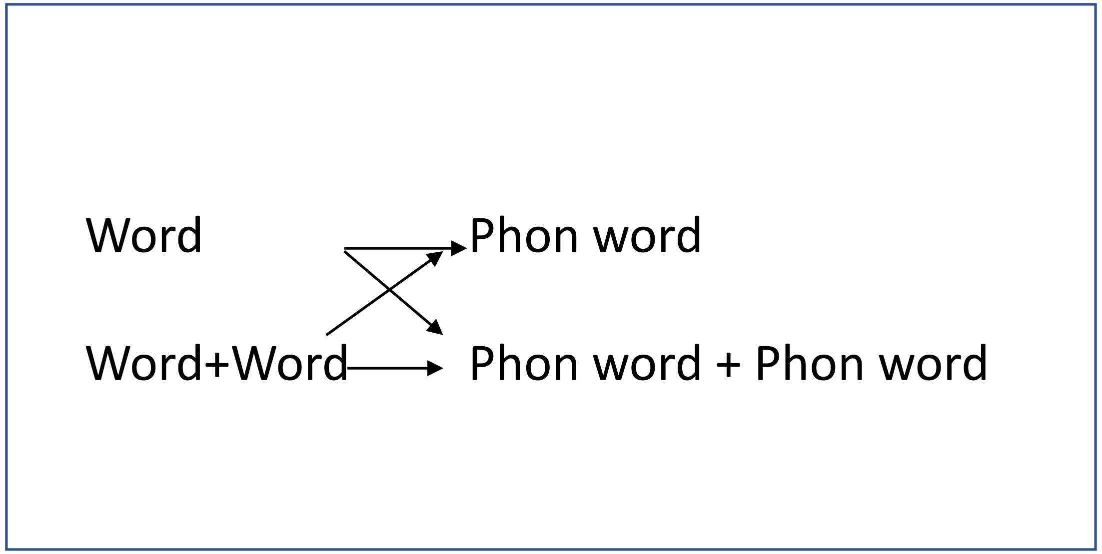

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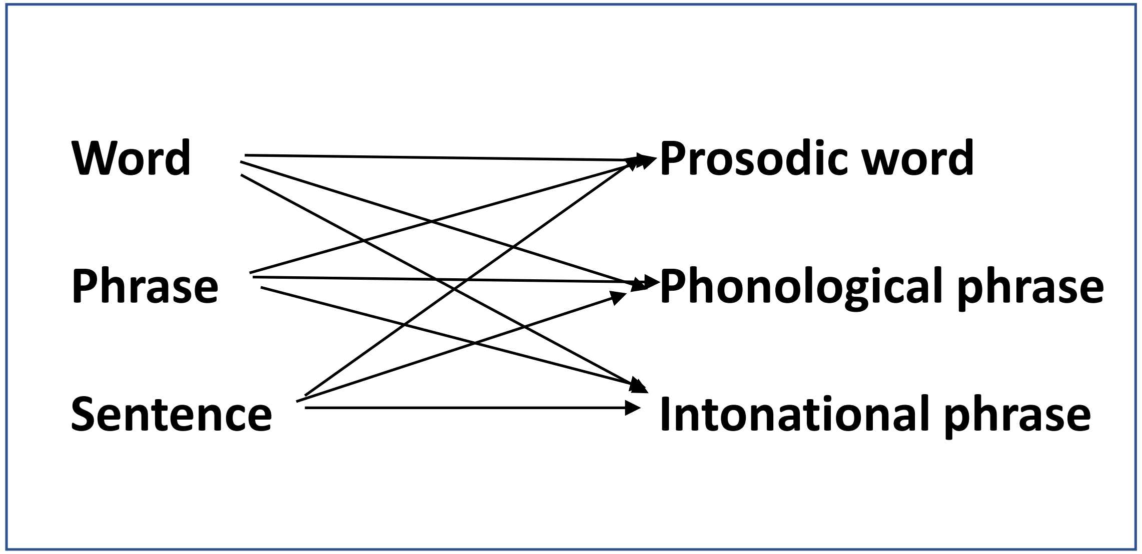
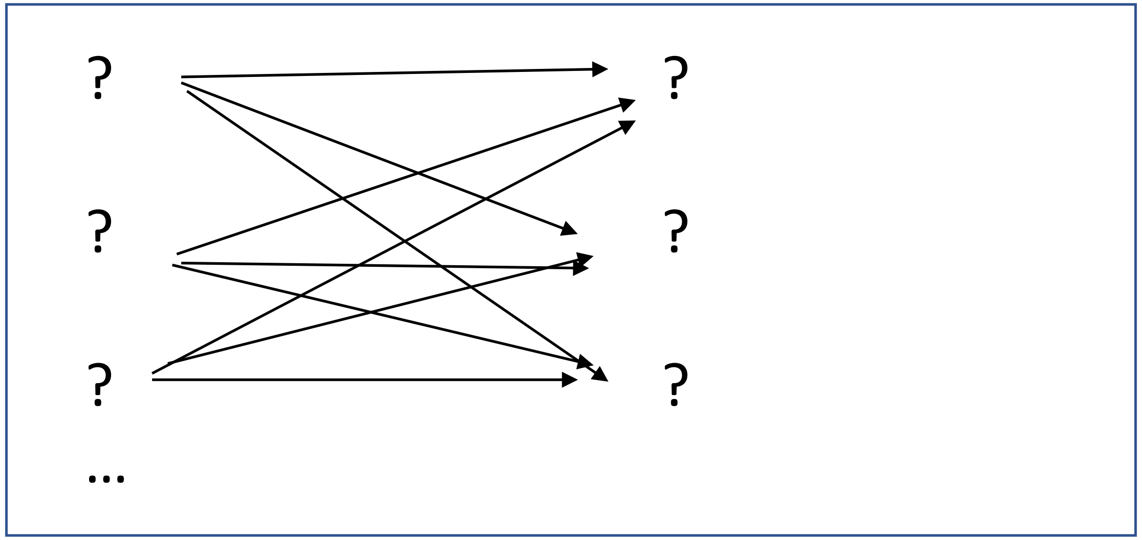


Reality



A bottom-up approach

Noninterruptability

Stress domain

Nonpermutability

Cluster reduction

Free Occurrence

?

Assimilation

Ciscategorial selection

Tone sandhi

Coordination

Something about clitics

Clitic

- What is a clitic?

Clitic

- What is a clitic?

- Something in between morphology and syntax

- Something that mixes canonical properties of affixes and canonical properties of words
  - Spencer and Luis (2012)

Clitic

- The traditional distinction

  - **Special clitic**: A morpheme that has a distribution that is not accounted for by the "normal" rules of the syntax.

  - **Simple clitic**: A morpheme that has a distribution of the "normal" rules of the syntax, but that is not a prosodic word.

  - Usually special clitics are not full prosodic words either, but Anderson (1992, 2005) assigns some "particles" to the class of special clitic just for having a non-normal distribution.

Special clitic

Juan	miró	a su	abuelo
John	saw	at his	grandfather

‘Juan saw his grandfather’

Juan	**le**	miró
Juan	**3SG**	saw

‘Juan saw him’

Special clitic

- Apart from pronouns in Romance (and other languages with similar differences between full NPs and pronouns) it remains mostly unclear what “normal” syntax refers to or how one is supposed to determine when syntax is normal.
  - Tallman 2018

- Despite the widely adopted distinction between simple and special clitics, there is little agreement on how to account for clitics theoretically.

Words, clitics and affixes

- The criteria in the literature for distinguishing between affixes and clitics is more or less the same as that for classifying words.

- There are no criteria that uniquely identify clitics. In practical terms the term “clitic” just means some morpheme that mixes affix values and word values in relation to the diagnostics that distinguish between affixes and words.

Criteria

- Permutability: Affixes occur in a fixed order whereas words are can be variably ordered.
- Interruptability: Affixes cannot be interrupted from their base by words.
- Coordination: Affixes cannot have wide scope under coordination, whereas words can.
- Prominence: Affixes do not project a stress domain whereas words do.
- Irregularities : affixes are subject to irregular morphophonological rules, but words are not.

Criteria (Tallman 2018)

- An important distinction

- **Practical category**: A category or even notion that is useful for organizing a descriptive grammar or for discussion in certain circumstances (e.g. “word” based on a single criterion)

- **Motivated category**: A category that has a number of converging properties that are logically independent and are not spurious (a chance correlation)

Variables (Tallman 2018)

- Binary: yes or no
- Ordinal: criteria has an implicational structure
- Continuous: In principle can vary from 0 to infinite

Variable	Type
Minimality domain with lexical root	ordinal
H-tone blocking	binary
Long distance tone reduction	binary
Conscription in reduplication	ordinal
Phonemic size	continuous
Segmental allomorphs	continuous
Suprasegmental allomorphs	continuous
Suppletive allomorphy	binary
Freedom	binary
Interruptability	ordinal
Permutation	ordinal
Scope over coordinated clauses	ordinal
Extraction to second position	binary
Auxiliary(-like) function	binary
Clause-final position	binary

Interruption

- In Chácobo there are two interruption tests dependening on whether the interrupting element is a “phrase” or a “particle”

(16) a. atʃ-a =má =kɨ
grab-TR =CAUS =DEC:P
‘He/she made him/her/it grab him/her/it.’

-ma
Interruptable = yes

-a
Interruptable = no

b. atʃ-a **yáma** =má =kɨ
grab-TR NEG =CAUS =DEC:P
‘He/she/it did**n’t** make him/her/it grab him/her/it.’

c. atʃ-a=ma **hóni siri**=́ wa=kɨ
grab-TR=CAUS **man old**=ERG TR=DEC:P
‘The **old man** made him/her/it grab him/her/it.’

Conscription in reduplication

- Elements that are copied when their base is copied are more affix like.

(12) [í.nà.tá.nɨ̀ʂ.tá.nɨ̀ʂ.kɨ̀.hó.nì]

ína	**tá**-nɨʂ=´	**tá**-nɨʂ=´	=kɨ	hóni
dog	foot-tie=LNK	foot-tie=LNK	=DEC:P	man

'The man has been tying the legs of dogs.'

Conscription in reduplication

- I call this property *conscription*; there are four patterns.

a. **Obligatorily conscripting**: If the affix or clitic is on the base, it must also appear on the reduplicant.

b. **Variably conscriptable**: If the affix of clitic is in the base it must appear on the reduplicant if not doing so would result in a monomoraic reduplicant. Otherwise the affix is freely conscriptable.

c. **Freely conscriptable**: If the affix or clitic is in the base, it *can* appear on the reduplicant, as long as the reduplicant corresponds to a contiguous string of morphemes found in the base.

d. **Unconscriptable**: The affix or clitic never appears on the reduplicant.

Conscription in reduplication

- Conscription level is correlates with boundedness and interruptability, but the relationship is not perfect.

- This is generally true of all the correlates

Overview

- Over nonlexical closed class morphemes coded, about 15 different classes emerge in the Chácobo case.
- Perhaps statistical prevalence can help us: K-means clusters provides 9 clusters …
- How do these categories fit into traditional categories presupposed in the literature?

Conclusions

- There is no evidence that clitics are motivated categories.

- The fact that one can partition morphemes into classes and assign them labels present in the literature is not evidence for the existence of such categories in the language itself let alone evidence that they are relevant for cross-linguistic study.

- Clitics are likely just epiphenomena that emerge from the way a specific linguist imposes the morphology syntax distinction in a specific language (which is heavily related to orthography)

Cross-linguistic considerations

- Is this problem really so bad?

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- Maybe.

Cross-linguistic considerations

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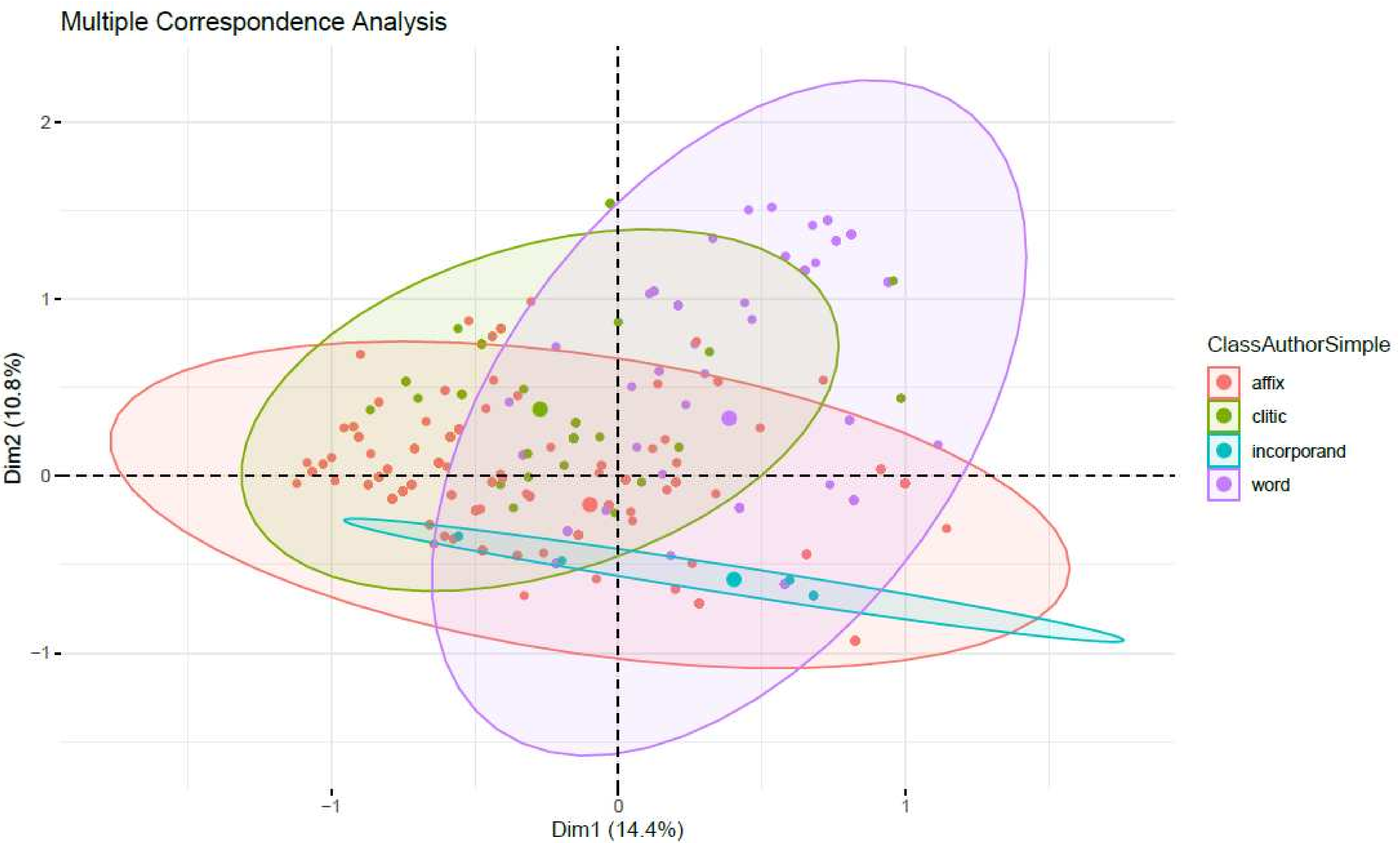
- Maybe.

- Consider the MCA analysis (done by Marc Tang) of over a database with approximately ~1150 morphemes coded for “word”, “affix”, “clitic” and “incorporand” following author classifications across 9 languages/grammars.


Multiple Correspondence Analysis
2
1
0
−1
−1
0
1
Dim2 (10.8%)
Dim1 (14.4%)
ClassAuthorSimple
affix
clitic
incorporand
word

What to do?

- The solution, I think, is to go probabilistic and focus on the diagnostics and their (non)correlations in individual languages and cross-linguistically.

- There will be a workshop on March 6th with Marc Tang and Dan Dediu that introduces R using it on data that are structured in a way similar to the Chacobo data just presented (but with more languages)

- The methods can help you see the relationship between different wordhood variables in your data that is hard to see by observing individual examples one by one.

What to apply constituency diagnostics to

- In the previous lecture (and we will return to this here) we talked about applying diagnostics to spans along a planar structure.

- In this perspective constituency diagnostics are applied over abstract linearly organized morphosyntactic positions.

- Another perspective (that we just reviewed) interprets the diagnostics with respect to individual morphemes (or more precisely base+morpheme pairs)

What to apply constituency diagnostics to

- Element level approach: code wordhood/constituency properties of individual base+morpheme pairs across the grammar.

- Position-span level approach: code spans of structure identified by constituency diagnostics.

- The difference between considering constituency diagnostics in terms of abstract morphosyntactic positions and of looking at them in terms of individual morphemes is just another way that constituency/wordhood diagnostics are ambiguous.

Constituency or wordhood criterial variables

Constituency test/diagnostic

- What’s a constituency diagnostic: in principle, any generalization across constructions that refers to spans of structure.

“I define a **constituency test** as a generalization within or across constructions that targets or crucially refers to some subspan of a planar structure. A constituency test can only be applied in a given language if it is specific enough such that it refers to a *well-defined* subspan. A subspan is **well-defined** if it contains a single left-edge (e.g. position 3) and single right-edge (e.g. position 8). The generalization on which the constituency test result is based should refer to syntagmatic properties that are frequently found in the literature on wordhood and constituency (non-interruptability, free occurrence, permutability) or phonological rules or constraints whose correct definition requires reference to boundaries in morphosyntactic structure (e.g. the application of a tone sandhi rule)” (Tallman 2018)

Constituency tests

- We can make a distinction between three types of constituency diagnostics.

    - Discrete/fracturable span identifying variables

    - Inherently stochastic variables

    - Locality variables

Discrete/fracturable span identifying variables

- Discrete/fracturable span identifying variables are variables that can identify spans as long as details are added to remove ambiguity in their application.

- We are going to primarily be talking about these type of constituency variables, but there are two other types that merit mention.

Inherently stochastic variables

- Inherently stochastic variables are those that cannot be stated in terms of generalizations over the planar structures, but would require a psycholinguistic experiment or an aggregation of corpus data in order to implement.

- We will not talk about these in this seminar, but they should be kept in mind for future research.

Constituency variables

- Suprasegmental domain
- Dissimilation
- Local
- Glottal dissimilation
- Teotitlan del Valle Zapotec glottal dissimilation

Abstract

Language specific

Constituency variables

- Suprasegmental domain
- Dissimilation
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- Teotitlan del Valle Zapotec glottal dissimilation

Abstract

Language specific

At what level on this continuum have we passed the threshold to become a comparative concept?

Well-defined

- A span is well-defined if it identifies a single left edge and right edge in the planar structure.
- Most of the constituency diagnostics do not identify spans as well-defined by themselves.

Fracturing

- The variables will often be stated in such a way that it is unclear how to apply them.

- In practice researchers add ad-hoc conditions to constituency diagnostics so that they can be used in some mostly unambiguous fashion.

- Often these ad-hoc conditions are unstated.

Fracturing

- You can't really blame descriptivists or anyone for doing this. When people make some statement about a diagnostic the statement is always extremely vague and ambiguous.

A **grammatical word** consists of a number of grammatical elements which:

(a) always occur together, rather than scattered through the clause (the criterion of cohesiveness);
(b) occur in a fixed order;
(c) have a conventionalised coherence and meaning.

Fracturing

- The fracturing method just states:
- Instead of adding just one ad-hoc stipulation to the test add all the ad-hoc stipulations that you can think of fracturing the test into subtests and report all the results.
- In a sense, test fracturing is already implicit in linguistic analysis, we are just making it less biased, by asking the linguist to report all the fractures rather than just the one that they find congenial to their argument.

Fracturing

- For example, the domain of noninterruptability refers to a span of structure that cannot be interrupted by another word.
- But what independent criteria do we use to identify words to begin with?
- We get different results depending on whether the interrupting element is a free form, some combination of free forms, some selectionally free element (clitic) etc.

Fracturing

- Three types of test fracturing
  - Minimal-Maximal fracturing: The diagnostic or domain can be defined based on positive evidence (minimal) or the absence of negative evidence (maximal).
  - Cross-linguistic fracture: The diagnostic or domain can be split into subtypes based on details that involve comparative concepts (concepts that can be observed across all languages).
  - Language-specific fracture: The diagnostic or domain can be split into subtypes based on details that involve language specific concepts.

Fracturing

- Three types of test fracturing

  - Minimal-Maximal fracturing: The diagnostic or domain can be defined based on positive evidence (minimal) or the absence of negative evidence (maximal).

Are these really different?

  - Cross-linguistic fracture: The diagnostic or domain can be split into subtypes based on details that involve comparative concepts (concepts that can be observed across all languages).

  - Language-specific fracture: The diagnostic or domain can be split into subtypes based on details that involve language specific concepts.

Constituency variables

- Free Occurrence
- Nonpermutability
- Noninterruptability
- Ciscategorial selection
- Subspan repetition
- Nonlocal Phonology
- Local phonology
- Exponence complexity
- …

Nonpermutability

- Elements of a word constituency occur in a fixed order
  - *Dixon & Aikhenvald 2002: 19; Aronoff & Fudeman 2005: 38*

- This is obviously a matter of degree language internally (and cross—linguistically?) because syntactic elements are frequently understood as occurring in fixed orders.

- *Furthermore,* variable affix ordering is a well attested phenomena
  - *Mattissen 2004; Rice 2011*

Nonpermutability

- There are two formulations of nonpermutability in the literature.
- One requires that all the elements in a span are fixed.
- Anther allows them to variably order but there must be a difference in scope.

Nonpermutability

“**Ordering:** Affix order within the word is quite rigid, while word order within phrases can vary. The degree of variation differs from language to language, of course, but virtually all languages allow some alternative word orders corresponding to the same essential meaning. In contrast, any variation we find in the order of word-internal affixes is virtually always correlated with a difference in sense.” (Anderson 2005: 9)

“Concerning criterion (b), it is in fact sometimes possible for affixes to occur in alternative ordering within a word, but there must then be a difference in meaning …” (Dixon and Aikhenvald 2002: 20)

Nonpermutability

- And even then, it’s a matter of degree. –There are many languiages that allow variable ordering of so-called ‘affixes’ without a noticeable difference in meaning.
    - Chintang (Bickel et al. 2007); Murrinpatha (2015)

Nonpermutability

Syntactic

Variable ordering (pragmatic difference)

Variable ordering (obligatory scope difference)
("Flexible nonpermutability")

Fixed ordering
("Strict nonpermutability")

Morphotactic

Nonpermutability

- A distinction can be made between strict (rigid order) and a flexible (strict order or scopal order) nonpermutability domains.

- This distinction reoccurs in language after language and seems to not be highly dependent on language-specific notions. (as far as I can tell!)

Position	Type	Elements
1	Slot	Connector
2	Zone	Noun Phrase (A/S/P),
3	Slot	*tsi*
4	Slot	Evidential/Modal
5	Slot	Affective
6	Slot	Noun Phrase (A/S), Noun
7	Slot	Body-part/locative prefix
8	Slot	**Verb Root**
9	Slot	Valence 1 (Transitivity)
10	Slot	Valence 2 (Antipassive *-mís*,
11	Zone	Valence 3 (Passive *-ʔaka*,
12	Zone	Valence 4 (interactional
13	Slot	*=yó*
14	Slot	Associated Motion
15	Slot	*=tápi*
16	Zone	Aspect:Durative
17	zone	Adverbial/modal
18	Slot	*tsi*
19	Slot	Evidential/Modal
20	Zone	Noun Phrase (A/S),
21	Slot	*WA* (TRANSITIVE)
22	Slot	*=KAN* (THIRD PERSON PLURAL),
23	Slot	TEMPORAL DISTANCE
24	Slot	**CLAUSE RANK AND TYPE**
25	Slot	EVIDENTIAL / MODAL
26	Slot	NOUN PHRASE (S/A)

Strict

(3) 6 8 10 12 24 26

*bakɨ̂ taa***-mis =má***=ki hóni*

boy give**-ANTIPASS =CAUS** =DEC:P man

'The man always sends the boy to give (out gifts or chicha, etc…).'

(4) 6 8 12 10 24 26

bakɨ̂ taa*=ma-mís** *=ki hóni*

boy give**=CAUS -ANTIPASS** =DEC:P man

'Intended: the man always sends the boy to give (out gifts or chicha (type of drink), etc.).'

Strict nonpermutability

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16	Zone	Aspect:Durative
17	zone	Adverbial/modal
18	Slot	*tsi*
19	Slot	Evidential/Modal
20	Zone	Noun Phrase (A/S),
21	Slot	*wa* (transitive)
22	Slot	*=kan* (third person plural),
23	Slot	Temporal distance
24	Slot	**Clause rank and type**
25	Slot	Evidential / Modal
26	Slot	Noun Phrase (S/A)

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5	Slot	Affective
6	Slot	Noun Phrase (A/S), Noun
7	Slot	Body-part/locative prefix
8	Slot	**Verb Root**
9	Slot	Valence 1 (Transitivity)
10	Slot	Valence 2 (Antipassive *-mís*,
11	Zone	Valence 3 (Passive *-ʔaka*,
12	Zone	Valence 4 (interactional
13	Slot	*=yó*
14	Slot	Associated Motion
15	Slot	*=tápi*
16	Zone	Aspect:Durative
17	zone	Adverbial/modal
18	Slot	*tsi*
19	Slot	Evidential/Modal
20	Zone	Noun Phrase (A/S),
21	Slot	*wa* (transitive)
22	Slot	=*kan* (third person plural),
23	Slot	Temporal distance
24	Slot	**Clause rank and type**
25	Slot	Evidential / Modal
26	Slot	Noun Phrase (S/A)

Flexible nonpermutability

(5) 8 [12] 24

haba ***=ma*** ***=bɨkɨ́*** *=kɨ*

run **=caus** **=intrc** =dec:p

'They (in cooperation) made him run.'

(6) 8 [12] 24

haba ***=bɨki*** ***=má*** *=kɨ*

run **=INTRC** **=CAUS** =DEC:P

'He made them run together (in cooperation).'

Nonpermutability

- In South Bolivian Quechua a distinction can be made between lexical and productive suffixes.
- Lexical suffixes are listemes with their root, and productive suffixes can occur on any root.
- But, there are some affixes that have lexical and productive versions which express comparable (or perhaps identical concepts).
  - Camacho-Rios (2019)

Nonpermutability

(7) *Maria Juan-man wak'a-ta riku***chi***-nqa*

Maria Juan-to sacred place-ACC show-3SG.FUT

'Maria will show the sacred place to Juan' (Constructed)

(8) *Maria Juan-man wak'a-ta riku***chi***-***chi***-nqa*

Maria Juan-to sacred place-ACC show-**CAUS**-3SG.FUT

'Maria will make someone to show the sacred place to Juan' (Constructed, Camacho-Rios 2019)

#	Type	Elements
1	slot	NP
2-8	slots	clitics 2-8 = 25-31
9	slot	NP
10	slot	verb base
11	slot	lexemic suffixes ya, ra, cha, raya, kipa, paya, **chiy (causative),** ri,
12	slot	naya, ra, rqaya, ykacha
13	slot	yu~yku
14	slot	ri
15	zone	rpa, ysi
16	slot	rqu
17	slot	**chi (causative)**
18	slot	naku, ku
19	slot	mu
20	slot	pu
21	slot	sha
22	slot	lla
23	slot	rqa, qti
24	slot	Person & number
25	slot	=manta
26	slot	(=ña)
27	slot	(=raq)
28	slot	(=puni) / (=taq)
29	slot	(=sina) / (=pis)
30	slot	(=qa), (=chus), (=chu), (=cha)
31	slot	(=ri)
32	slot	NP

Nonpermutability

- Only a subset of verbs in SB Quechua can receive two causative morphemes.
- And when this occurs, it is easy to show that the first causative cannot be interrupted by anything, whereas the second causative can be.
- Are the –chi's different morphemes?

Nonpermutability

- If we assume they are the same morpheme then the domain of strict nonpermutability is 10-10.

#	Type	Elements
1	slot	NP
2-8	slots	clitics 2-8 = 25-31
9	slot	NP
10	slot	verb base
11	slot	lexemic suffixes ya, ra, cha, raya, kipa, paya, **chiy (causative),** ri,
12	slot	naya, ra, rqaya, ykacha
13	slot	yu~yku
14	slot	ri
15	zone	rpa, ysi
16	slot	rqu
17	slot	**chi (causative)**
18	slot	naku, ku
19	slot	mu
20	slot	pu
21	slot	sha
22	slot	lla
23	slot	rqa, qti
24	slot	Person & number
25	slot	=manta
26	slot	(=ña)
27	slot	(=raq)
28	slot	(=puni) / (=taq)
29	slot	(=sina) / (=pis)
30	slot	(=qa), (=chus), (=chu), (=cha)
31	slot	(=ri)
32	slot	NP

#	Type	Elements
1	slot	NP
2-8	slots	clitics 2-8 = 25-31
9	slot	NP
10	slot	verb base
11	slot	lexemic suffixes ya, ra, cha, raya, kipa, paya, **chiy (causative),** ri,
12	slot	naya, ra, rqaya, ykacha
13	slot	yu~yku
14	slot	ri
15	zone	rpa, ysi
16	slot	rqu
17	slot	**chi (causative)**
18	slot	naku, ku
19	slot	mu
20	slot	pu
21	slot	sha
22	slot	lla
23	slot	rqa, qti
24	slot	Person & number
25	slot	=manta
26	slot	(=ña)
27	slot	(=raq)
28	slot	(=puni) / (=taq)
29	slot	(=sina) / (=pis)
30	slot	(=qa), (=chus), (=chu), (=cha)
31	slot	(=ri)
32	slot	NP

Nonpermutability

- If we assume they are the same morpheme then the domain of strict nonpermutability is 10-10.
- If we assume they are different, then the domain of strict nonpermutability is 10-16.

Nonpermutability

Position	Type	Elements
1	Slot	Noun Phrase (A/S), Conjunction, Coordinated group
2	Slot	Noun Phrase (O), Conjunction, Coordinated group
3	Slot	*Verb*
4	Slot	Motion verb
5	Slot	Positional verb
6	Slot	Imperative; motion
7	Slot	Aspect/Tense
8	Slot	Person/ S/A NP
9 10 11 12 13 14 15	Slot Slot Slot Slot Slot Slot Slot	**Person/DIST** **Neg** **Evidential, interrogative,** **Aspect, evidential** ***Ni*** **Topic** **Stative**

- In Naso (Chibchan) nonpermutability identifies a span with the whole NP object.
- This type of thing isn't too uncommon in the data and shows that linguists tend to be biased in their application of tests – they apply them in such a way such that it does not contradict the orthographic word.

Noninterruptability

- Words are supposed be non-interruptable by other words.
    - Bauer (2017)

Noninterruptability

- This formulation is circular
  - Mugdan 1996
  - (Brüening’s insistence that X^0 isn’t really a word doesn’t provide any insight by itself. Whatever X^0 identifies, we should try to reformulate it as a typological variable)

- Haspelmath (2011) suggests that the interrupting element should be a free form.

- Bauer’s (2017) discuss suggests that the interrupting element could be a (bound) clitic.

Noninterruptability

- For many cases of noun incorporation, the bare noun would be excluded from the verbal word on Haspelmath's (2011) view.
- Maybe the interrupting element should be phrase that has the possibility of containing more than one free form?

Noninterruptability

Syntactic

Interruptable by combination of free forms
(e.g. Noun phrase)

Interruptable by free form
(e.g. “particle)

Noninterruptable by promiscuous element
(e.g. “clitic”)

Morphotactic

Noninterruptable

Noninterruptability

Position	Type	Elements
1	slot	connector
2	slot	adverbial interrogatives
3	zone	Negation, NP(P), PP(Obl), AdvP, =no
4	slot	NP(A/S)
5	slot	Negation *unká*
6	slot	person index (A/S)
7	slot	verb core
8	slot	transitive (caus *-ta*, appl *-ñaa*, recip *-ka*)
9	slot	negation *-la*
10	slot	tense (past *-cha* , future *-je*)
11	slot	middle *=o*
12	slot	aspect (perfective *=mi*)
13	slot	emphatic *táa*
14	zone	mood, aspect (frustrative *jlá*, iterative *=no*)
15	slot	emphatic *=ko*
16	zone	NP(P), PP(Obl), AdvP, *=no*

- **In Yukuna the negative marker is a free form.**

(1) a. É pi=i'mi-chá-ká

INT2SG=go-PST-INT

'Did you go?'

b. Unká.

NEG

'No.'

(Lemus in prep)

Position	Type	Elements
1	Slot	Connector
2	Zone	Noun Phrase (A/S/P),
3	Slot	*tsi*
4	Slot	Evidential/Modal
5	Slot	Affective
6	Slot	Noun Phrase (A/S), Noun
7	Slot	Body-part/locative prefix
8	Slot	**Verb Root**
9	Slot	Valence 1 (Transitivity)
10	Slot	Valence 2 (Antipassive *-mís*,
11	Zone	Valence 3 (Passive *-ʔaka*,
12	Zone	Valence 4 (interactional
13	Slot	*=yó*
14	Slot	Associated Motion
15	Slot	*=tápi*
16	Zone	Aspect:Durative
17	zone	Adverbial/modal
18	Slot	*tsi*
19	Slot	Evidential/Modal
20	Zone	Noun Phrase (A/S),
21	Slot	*wa* (transitive)
22	Slot	*=kan* (third person plural),
23	Slot	Temporal distance
24	Slot	**Clause rank and type**
25	Slot	Evidential / Modal
26	Slot	Noun Phrase (S/A)

Noninterruptability

- This is also true of Chacobo, except the negation marker can occur in position 11, making the domain relatively smaller.

8-9 [11] 24
atʃ-a ***yáma*** *=má* *=kɨ*
grab-TR **NEG** =CAUS =DEC:P
'He/she/it did**n't** make him/her/it grab him/her/it.'

Noninterruptability

Position	Type	Elements
1	slot	connector
2	slot	adverbial interrogatives
3	zone	Negation, NP(P), PP(Obl), AdvP, =no
4	slot	NP(A/S)
5	slot	Negation *unká*
6	slot	person index (A/S)
7	slot	verb core
8	slot	transitive (caus -*ta*, appl -*ñaa*, recip -*ka*)
9	slot	negation -*la*
10	slot	tense (past -*cha* , future -*je*)
11	slot	middle =*o*
12	slot	aspect (perfective =*mi*)
13	slot	emphatic *táa*
14	zone	mood, aspect (frustrative *jlá*, iterative =*no*)
15	slot	emphatic =*ko*
16	zone	NP(P), PP(Obl), AdvP, =*no*

- The negative morpheme in Yukuna could be included in a noninterruptable span if the interrupting element is a combination of more than one free form (i.e. an NP

Position	TYPE	ELEMENTS
1	Slot	Connector
2	Zone	Noun Phrase (A/S/P),
3	Slot	*tsi*
4	Slot	Evidential/Modal
5	Slot	Affective
6	Slot	Noun Phrase (A/S), Noun
7	Slot	Body-part/locative prefix
8	Slot	**Verb Root**
9	Slot	Valence 1 (Transitivity)
10	Slot	Valence 2 (Antipassive *-mís*,
11	Zone	Valence 3 (Passive *-ʔaka*,
12	Zone	Valence 4 (interactional
13	Slot	*=yó*
14	Slot	Associated Motion
15	Slot	*=tápi*
16	Zone	Aspect:Durative
17	zone	Adverbial/modal
18	Slot	*tsi*
19	Slot	Evidential/Modal
20	Zone	Noun Phrase (A/S),
21	SLOT	*WA* (TRANSITIVE)
22	SLOT	*=KAN* (THIRD PERSON PLURAL),
23	SLOT	TEMPORAL DISTANCE
24	SLOT	**CLAUSE RANK AND TYPE**
25	SLOT	EVIDENTIAL / MODAL
26	SLOT	NOUN PHRASE (S/A)

Noninterruptability

- This is also true of Chacobo, except the negation marker can occur in position 11, making the domain relatively smaller.

(17) *atʃ-a=ma* ***hóni siri=´*** *wa =kɨ*
grab-TR=CAUS **man old**=ERG TR =DEC:P
'The **old man** made him/her/it grab him/her/it.'

Noninterruptability

- We also get different results if bound elements which can occur in multiple positions of the verbal planar structure are considered the interrupting elements.

- Some languages seem to like such elements (e.g. Hup) other languages do not (e.g. Teotitlan del Valle Zapotec).

Noninterruptability

(19)a) 7 22 24

*ʔãh himɨhɨn-yɨ́ʔ-ɨ́y=**cud***

1sg forget-TEL-DYNM=INFR

'I forgot it, apparently.'

b) 7 19 22

*ní-**cud**-úʔ ?*

be-INFR-INT

'(She's) there, huh?'

Position	Type	Elements
1	Zone	A/S/O/oblique
2	Slot	Proclitic
3	Zone	Prefix (interactional) / O (simplex)
4	Zone	Prefix (reflexive/passive) / (simplex)
5	Slot	Causative root
6	Slot	Prefix (factitive)
7	Zone	**Verb stem**
8	Slot	Telic
9	Slot	Venitive *-ʔay*
10	Slot	Applicative *-ʔũh*
11	Slot	Completive *-hũʔ*
12	Slot	Counterfactual *-tæ̃ʔ*
13	Slot	Emphasis
14	Zone	Perfective/emphasis
15	Slot	Clausal negation *-nɨh*
16	Slot	Habitual/distributive/future
17	Slot	Evidential/frustrative/repetitive
18	Slot	Inchoative
19	Slot	Inferred evidential *-ni*
20	Slot	Filler
21	Slot	**Boundary suffix (clause type marker)**
22	Slot	Counterfactual
23	Slot	Emphatic coordinator
24	Zone	Inferred/nonvisual evidential
25	Slot	Repetitive
26	Slot	Reportive evidential
27	Slot	Habitual/distributive
28	Slot	Frustrative
29	Slot	Emphasis
30	Slot	Contrast
31	Zone	Intensifier / adversative conjunction/ persistive / epistemic modality
32	Zone	A/S/O/oblique

O

Noninterruptability

- I = Free complex 2-31
- I = Free simplex 5-31
- I = emphasis 2-13
- I = habitual 2-16
- I = evidential 2-18
- I = declarative 2-20

Ciscategorial selection

‘Clitics can exhibit a low degree of selection with respect to their hosts, while affixes exhibit a high degree of selection with respect to their stems’ (Zwicky and Pullum 1983)

Ciscategorial selection

- There’s a a lot of terminology in the context of wordhood diagnostics which is highly ambiguous.

- ‘Attach’ sometimes refers to distributional constraints, sometimes to phonological ones, sometimes both and sometimes to orthographic representations only (sometimes it just means “occurs beside” in one context).

- ‘Select’ is similarly ambiguous, but the reason is more insidious, because things ‘select’ undefined or stipulated ‘constituents’ (or phrases?) or pieces of structure.

Ciscategorial selection

- The clearest use of ‘select’ is when we talk about semantic combination.
- In my view, an **element** in the planar structure is just sometimes that the verb base can be viewed as selecting (or vice versa) in this sense.
- This is why adjectives are not elements, but NPs are.

Ciscategorial selection

Position	Type	Elements
1	slot	connector
2	slot	adverbial interrogatives
3	zone	Negation, NP(P), PP(Obl), AdvP, =no
4	slot	NP(A/S)
5	slot	Negation *unká*
6	slot	person index (A/S)
7	slot	verb core
8	slot	transitive (caus *-ta*, appl *-ñaa*, recip *-ka*)
9	slot	negation *-la*
10	slot	tense (past *-cha*, future *-je*)
11	slot	middle *=o*
12	slot	aspect (perfective *=mi*)
13	slot	emphatic *táa*
14	zone	mood, aspect (frustrative *jlá*, iterative *=no*)
15	slot	emphatic *=ko*
16	zone	NP(P), PP(Obl), AdvP, *=no*

- The span of only ciscategorial elements in Yukuna is 7-10.
- These elements can only combine with the verb base.
- Middle markers and pronominal markers combine semantically with nouns.

Ciscategorial selection

- If selection is supposed to distinguish affixes from non-affixes and the former are less selectionally free, we have to pin down a relatively comparable notion of selection.

- Ciscategorial element: Elements that only combine with a single part of speech class.

- Transcategorial element: Elements that can combine with more than one part of speech class.

Ciscategorial selection

- But a domain of ciscategoriality would be ambiguous:

- Do we skip intervening transcategorial elements when we define the left and right edge?

Ciscategorial selection

- **Minimal**: The span overlapping the verb base that contains only V-ciscategorial elements.

- **Maximal**: The span overlapping thre verb base whose edges contain V-ciscategorial elements.

Subspan repetition

- Coordination is a traditional constituency diagnostic.
  - Chomsky 1957; Goodall 2017

- It tends to be applied in at least two ways depending on whether one is concerned with defining some span of structure or whether one is concerned with the wordhood status of individual elements.

Subspan repetition

- Coordination is a funny test in the literature: its simultaneously the most used test (Osborne 2018), but also the most criticized test (e.g. Carnie 2010) in the literature.

> "…the possibility of conjunction offers one of the best criteria for the iniial determination of phrase structure" (Chomsky 1957: 24)

> … this test [coordination] is prone to false positives. For example, it would appear as if the subjects and the verbs form constituents as distinct from the subject in the following right-node raising sentence… However, evidence from other constituency tests, such as movement or replacement suggests that the verb and the object form a constituent distinct from the subject" (Carnie 2010: 21)

Subspan repetition

- In Osborne’s survey of constituency diagnostics, he notes that coordination is the most permissive – being the only test that identifies “subphrasal” (between word and phrase) constituents consistently.

- Basically its difficult to assess the meaning of constituency diagnostics in the absence of the linguist’s assumptions about gapping and ellipsis.
  - Beavers and Sag 2004

Subspan repetition

- At the same time in the typological literature, it is recognized that there is not a clear cut distinction between coordination and subordination (depends on what criteria is chosen)
  - Cristofaro 2005; Bickel 2010, *inter alia*

- Actually… there's not a clear distinction between serialization and coordination (is serialization a type of coordination or not?)

- There is no clear distinction between verb compounding and serialization…

Subspan repetition

- A general term for all these “recursive” like processes is “subspan repetition”.
- If some subspan (which can be the whole template) is identified by coordination, then by the same logic subordination, serialization, verb compounding, auxiliation, also identify constituents.
- But it’s more complicated, because in every case we will need to split the relevant constructions according to the minimal-maximal fracturing logic.

Coordination

[Bruce loved] ~~tuna salad sandwiches~~ and [Dory hated] tuna salad sandwiches

Subspan repetition

- Diagnostic interpretation 1: Coordination involves repetition of a constituent in the sentence.

- Diagnostic interpretation 2: Clitics have wide-scope under coordination.
  - Spencer and Luis 2012

Subspan repetition

- Let’s start with interpretation 1.

- The most uncontroversial constituent identified by a given coordination construction is one that is the largest possible.

- To me, a gapping or ellipsis interpretation of missing material becomes dubious if the relevant categories just never surface in the sentence.

Table 1. Tentative planar template for English

#	Type	Elements
1	zone	Sentence
2	slot	NP
3	zone	Manner
4	slot	Modal
5	zone	Negative, quantifier (not, all, adv)
6	slot	Perfect (ha)
7	slot	Infl (-z,-v)
8	zone	Negative, quantifier (not, all, adv)
9	slot	Progressive (be)
10	slot	Infl (n)
11	zone	Negative, quantifier (not, all, adv)
12	slot	Passive (be)
13	slot	Infl (n)
14	zone	Negative, quantifier (not, all, adv)
15	slot	Derivation (un,de,re…)
16	slot	Latinate prefix (ex, in…)
17	slot	Verb
18	slot	Infl (ing, ed, en s)
19	slot	NP
20	slot	Compl (to)
21	slot	NP, S
22	slot	P (to, on, over)
23	slot	NP
24	zone	Adv, PP, S

Subspan repetition

- What would be identified for English “and” constuctions as the *largest* (Maximal) constituent.

- The entire template.

Table 1. Tentative planar template for English

#	Type	Elements
1	zone	Sentence
2	slot	NP
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8	zone	Negative, quantifier (not, all, adv)
9	slot	Progressive (be)
10	slot	Infl (n)
11	zone	Negative, quantifier (not, all, adv)
12	slot	Passive (be)
13	slot	Infl (n)
14	zone	Negative, quantifier (not, all, adv)
15	slot	Derivation (un,de,re...)
16	slot	Latinate prefix (ex, in...)
17	slot	Verb
18	slot	Infl (ing, ed, en s)
19	slot	NP
20	slot	Compl (to)
21	slot	NP, S
22	slot	P (to, on, over)
23	slot	NP
24	zone	Adv, PP, S

Subspan repetition

- But what we be identified as the constituent as nonfinite “to” sentences.
- It would be missing the subject, and modals.
- I want him to *should go.

Subspan repetition

- In Chácobo asyndetic coordination can be shown to repeat a span from 6 to 17.

(23)	6	8	9 [	17]	6	8	9	[	17]
	waka	*atʃ*	*-a*	*=tɨkɨ́(n)=kɨá*	*notí*	*nɨș*	*-a*		*=tɨkɨ(n)=kɨá*
	cow	grab	-tr	=again=cntrfct	canoe	tie	-tr		=again=cntrfct
	20	21		24					
	adaŋ=´	*=wa*		*=kɨ*					
	Adam=erg	=tr		=dec:p					

‘Adam attempted to grab the cow again and attempted to tie the canoe again.’

Subspan repetition

- It cannot really be shown that anything larger that this is repeated, outside of a full sentence.

- Full sentence repetition is a different construction in Chácobo that allows connectors like *hatsi* ´then´*hakirɨkɨ́* ‘after this’ with a wider array of semantic interpretations.

- While it is true that coordination is permissive and tricky to interpret because of ellipsis and gapping, the largest possible domain (maximal subspan repetition) provides a stricter interpretation that can be used to provide consistent cross-linguistic results.

Subspan repetition

- Interpretation 2: If affixes cannot display wide-scope under coordination, then we can define a domain of minimal (nonwidescope) subspan repetition as one where none of the elements flanking the verb base can display wide scope in the relevant construction.

- To test this, you find categories/morphemes that can repeat in both spans called on by the subspan repetition construction, and see if one of them can be dropped while the other is interpreted as modifying both parts of the linked/repeated spans.

Subspan repetition

(24) Jonah took the class again and wrote his essay again =

(25) Jonah took the class ~~again~~ and wrote his essay again

Subspan repetition

(24) Jonah took the class again and wrote his essay again =

(25) Jonah took the class ~~again~~ and wrote his essay again

'again' has widescope, because this sentence can be interpreted as meaning that Jonah is taking the class for at least the second time.

Subspan repetition

(26) Jonah retook the class and rewrote his essay again ≠

(27) Jonah ~~re~~took the class and rewrote his essay again

Subspan repetition

(26) Jonah retook the class and rewrote his essay ≠

(27) Jonah ~~re~~took the class and rewrote his essay

‘re-’ cannot have widescope, because this cannot be understood (at least not without a lot of imagination?) as meaning that John took the class a second time

Phonological tests

- Many domains for phonological tests are ambiguous in a sense that can be interpreted in the context of a minimal-maximal logic.

- This can be illustrated with glottal dissimilation with Teotitlan del Valle Zapotec.

Glottal dissimilation

- Minimal domain: identifies a span of 12-14: every position contains elements that have a glottal stop such that one can observe positive evidence for the process.

(19) rutaˈwǣ’nānbā
10-12-14=22=26
ru-**ta’w-æ’**n=ān=bā
HAB-sell-DIM=3sg.f=then
‘So he sells (which is good)’

(20) *bata’w xhlyā’tæ̰̂n gû’n*
10-12 15=16=22 24
ba-**ta’w xhlyā’**=tæ̰=angû’n
COMPL-sell in.vain=intsf=3sg.ifbull
‘He sold the bull in vain.’

Glottal dissimilation

- But no prefixes (position 10) contain glottal stops so one cannot tell whether the process applies or not. This is why we introduce the maximal domain. It extends the minimal domain until we have overt negative evidence against glottal dissimilation applying. So the maximal domain is 3-14.

(21) *lâʼ dǽʼ rûʼkán*
[2] 12 25
lâʼ 0-dǽʼ ruʼkán
Huaje POT-be.picked mouth.dem
‘Huaje will be picked at that border’

Glottal dissimilation

- Minimal domain (12-14)
  - Where you can see the process applying
- Maximal domain (3-14)
  - Where you have no evidence against its application / where it could be applying vacuously.

Positions	Type	Elements
1	zone	subordinators
2	zone (focus position)	NP {A,S,P, }/ Adv.P
3	slot	Focus marker (=ēn)
4	slot	adv á= 'already'
5	slot	Clausal Negator (kēd=)
6	zone	=pkā, =kā, =zī (adverbs of frequency and manner)
7	slot	=zá (adverb of equality)
8	slot	=rú (adverb of comparison)
9	slot	= pron.indf tū= / xī= /kalī= /gūl= imp?
10	slot	Aspect
11	slot	Motion
12	slot	Verb base
13	slot	Comitative
14	slot	affective
15	zone	(xhlyā / dxī) Manner adverbs
16	slot	Intensifier (some are free forms)
17	slot	Negation (=di)
18	zone	adverbs of frequency and manner
19	slot	=zá (adverb of equality)
20	slot	=rú (adverb of comparison)
21	slot	Reciprocal
22	slot /zone?	Pronoun/NP {A,S}
23	slot	Pronoun/NP {P}
24	slot	adverb
25	slot	subordinate clause
26	zone	=bā, =x Discourse elements

Phonological diagnostics

- There are lots of other phonological tests depending on the phonology / morphophonology of the language.

  - Minimal stress domain: the span of positions where you can see stress occurring

  - Culminative Maximal stress domain: the span of positions where stress is necessarily culminative (just one)

  - Obligatory Maximal stress domain: the span of positions where stress is obligatory.

Typology of constituency and convergence

Typology of constituency and convergence

- Results can be coded in a database so that cross-linguistic variation in constituency diagnostic (non)convergence can be measured.
- One difficulty is how to code constituency test fracturing in a consistent manner (something we are continuing to work on)

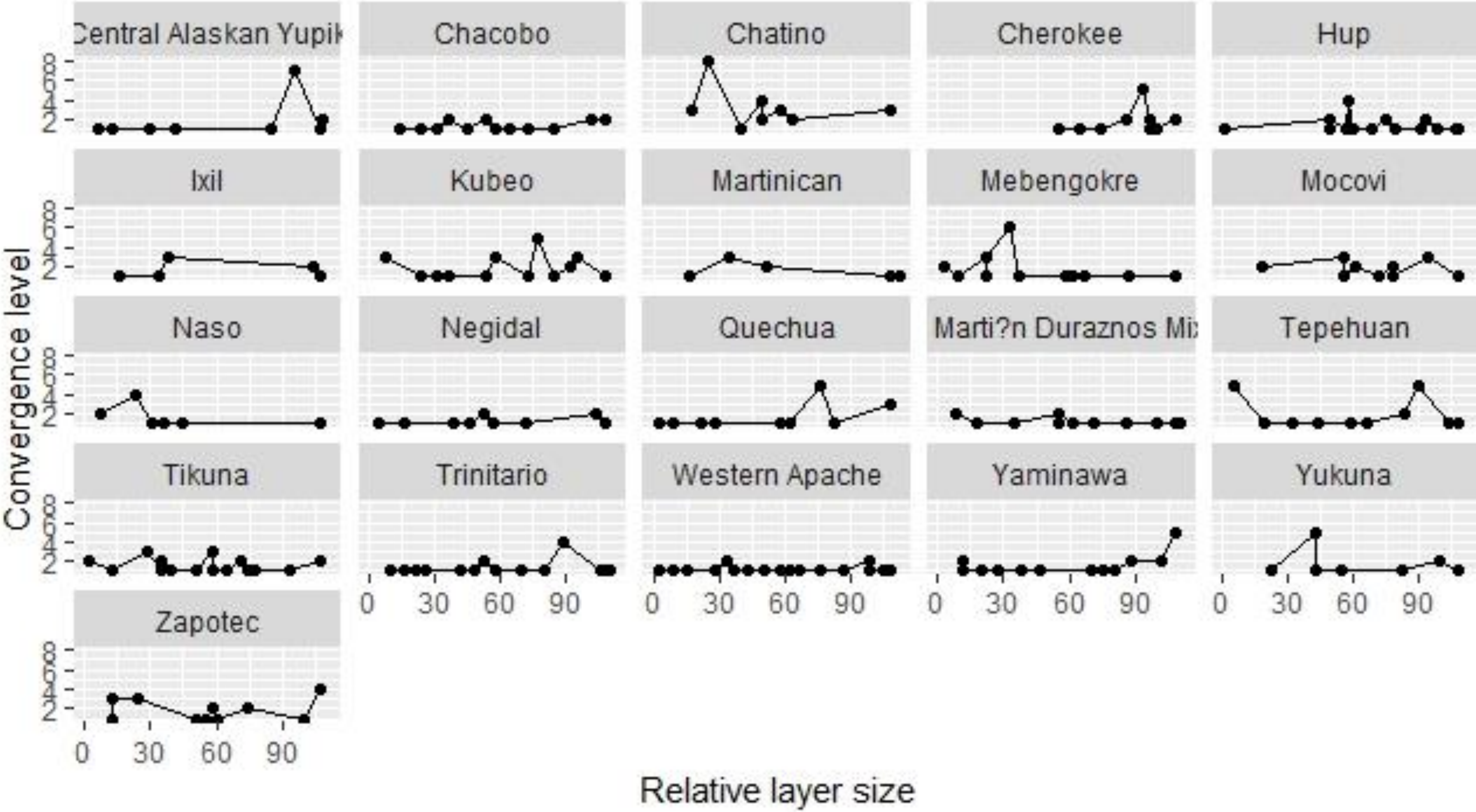
1	Grouping_	Class	Modular t	Cross lang	Minimal	Langu	Overl:	Overl:	Language	Left.span	Right.spar	Size	Converge	Largest	Relative	Positions_
95	moc13_Fr	Free Occu	Indeterminate		Minimal	A/S=2	V	4	Mocovi	3	10	8	1	13	0.6154	13
96	moc5_Sel	Ciscategoi	Morphosyntactic				V	4	Mocovi	4	12	9	2	13	0.6923	13
97	moc8_pal	Segmenta	Phonological				V	4	Mocovi	4	12	9	2	13	0.6923	13
98	moc16_De	Exponenc	Indeterminate				V	4	Mocovi	3	11	9	1	13	0.6923	13
99	moc3_Inte	Noninterr	Morphosy	Free Complex			V	4	Mocovi	2	12	11	3	13	0.8462	13
100	moc10_Str	Suprasegr	Phonological				V	4	Mocovi	2	12	11	3	13	0.8462	13
101	moc14_Fr	Free Occu	Morphosyntactic		Maximal		V	4	Mocovi	2	12	11	3	13	0.8462	13
102	moc9_len	Segmenta	Phonological				V	4	Mocovi	1	13	13	1	13	1	13
103	nas04_ser	Subspan r	Morphosyntactic			Serial	V		Naso	5	5	1	2	16	0.0625	16
104	nas05_stre	Suprasegr	Phonological				V		Naso	5	5	1	2	16	0.0625	16
105	nas01_Fre	Free Occu	Morphosyntactic				V		Naso	5	7	3	4	16	0.1875	16
106	nas03_Inte	Noninterr	Morphosyntactic				V		Naso	5	7	3	4	16	0.1875	16
107	nas06_De	Exponenc	Indeterminate				V		Naso	5	7	3	4	16	0.1875	16
108	nas07_De	Exponenc	Indeterminate				V		Naso	5	7	3	4	16	0.1875	16
109	nas09_Fix	Nonpermi	Morphosy	Strict			V		Naso	2	5	4	1	16	0.25	16
110	nas02_Inte	Noninterr	Morphosyntactic				V		Naso	3	7	5	1	16	0.3125	16
111	nas10_Fix	Nonpermi	Morphosy	Flexible			V		Naso	2	7	6	1	16	0.375	16
112	nas08_gac	Subspan r	Morphosyntactic			Coord	V		Naso	1	16	16	1	16	1	16
113	tla04_Seri	Subspan r	Morphosyntactic		Minimal	Serial	V	6	Tepehuan	6	6	1	5	19	0.0526	19
114	tla10_Red	Subspan r	Morphosyntactic			Redu	V	6	Tepehuan	6	6	1	5	19	0.0526	19
115	tla12_Seg	Segmenta	Phonological			Subtr	V	6	Tepehuan	6	6	1	5	19	0.0526	19
116	tla15_Sres	Suprasegr	Phonological			Stress	V	6	Tepehuan	6	6	1	5	19	0.0526	19
117	tla19_Free	Free Occu	Morphosyntactic		Minimal	Minin	V	6	Tepehuan	6	6	1	5	19	0.0526	19
118	tla13_Dev	Exponenc	Indeterminate			Alterr	V	6	Tepehuan	6	8	3	1	19	0.1579	19
119	tla18_Syll	Suprasegr	Phonological			Const	V	6	Tepehuan	2	6	5	1	19	0.2632	19

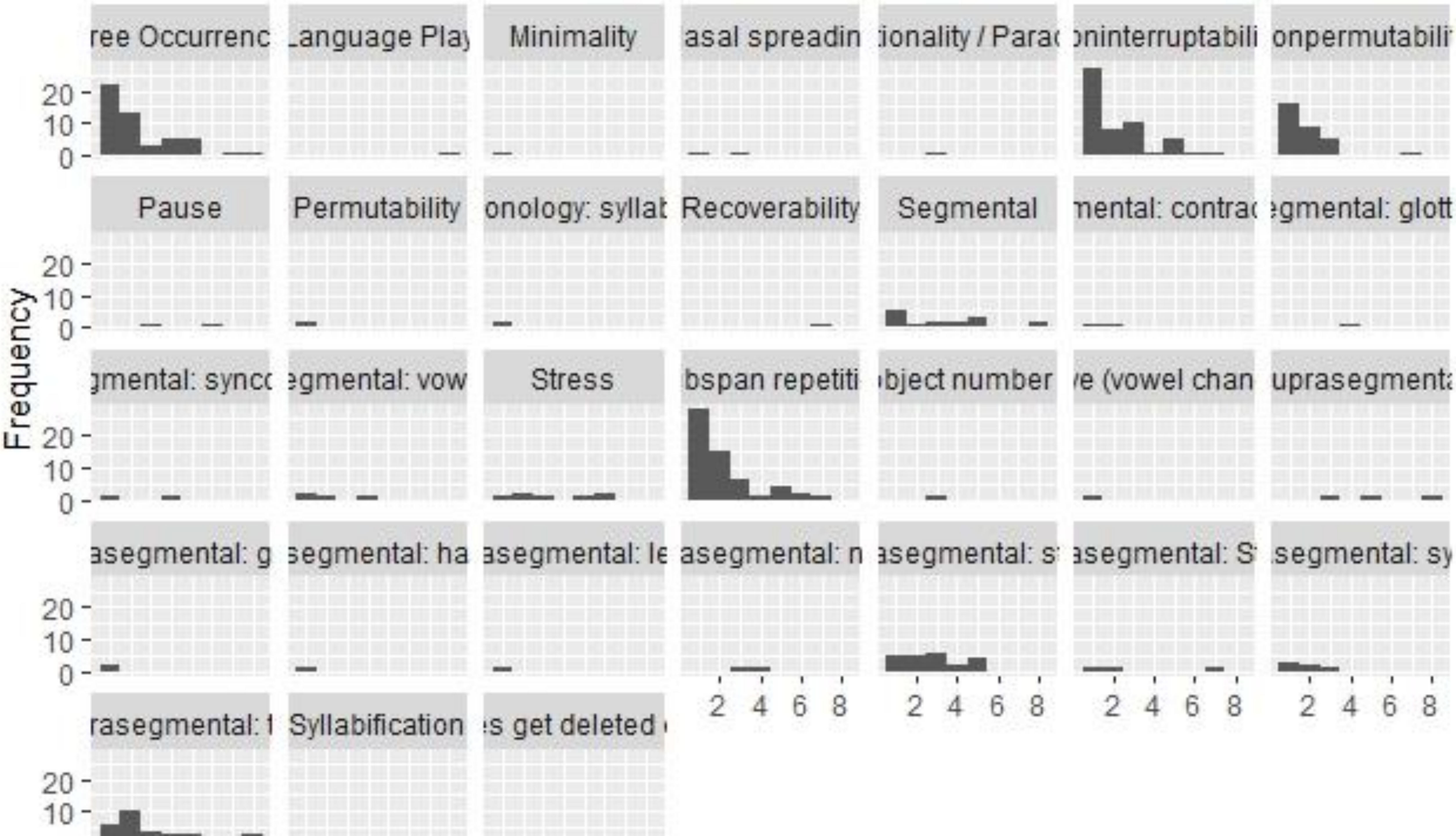
Typology of constituency and convergence

- With the diagnostics coded across the languages we can observe cross-linguistic variation with respect to convergence in constituency.

- Languages vary in the extent to which ‘words’ are motivated constituents and how large these converging constituents are.
  - Relative layer size = Size of span identified by diagnostic / size of largest span identified by any diagnostic in the language

- Constituency diagnostics vary in terms of their comparability (whether they apply to all languages) and how well they predict convergences (some diagnostics are better than others


Central Alaskan Yupik
Chacobo
Chatino
Cherokee
Hup
Ixil
Kubeo
Martinican
Mebengokre
Mocovi
Naso
Negidal
Quechua
Marti?n Duraznos Mi
Tepehuan
Tikuna
Trinitario
Western Apache
Yaminawa
Yukuna
Zapotec
Convergence level
Relative layer size
0
30
60
90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Frequency
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10
20
2
4
6
8

Another question: Convergence beyond chance

- So we have very weak convergences in some languages.

- How do we know those convergences are not a matter of chance?

- See Tallman (*forthcoming*)